

TEAM BRIEFING · SIMPLE ENGLISH

Rai Mai Phao
Zero-Burn Farm

Farmers burn crop waste because it is **the cheapest way**. We want to make crop waste **worth money**. Our machine **KhumKha** ("worth it" in Thai) sits in a **village station**. It makes animal food, fuel blocks or compost. Farmers sell with **LINE**, and a **village driver** picks up the waste, like Grab.

26.0 M	~1 ฿	KhumKha	~200k ฿
rai burned in Thailand in 2026 (a record)	we pay for 1 kg of dry crop waste	"worth it": one machine, 3 products	for one machine · pay back in about 17 months

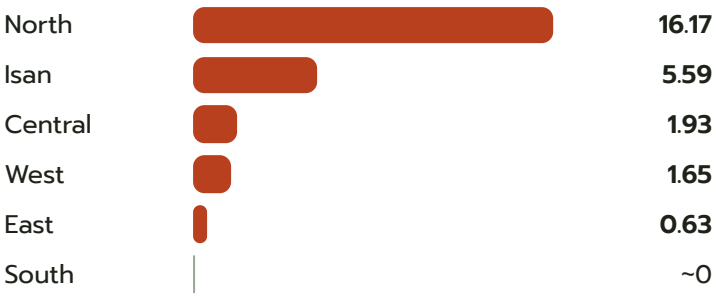
SDG 3 Health · SDG 8 Good jobs · SDG 12 No waste · SDG 13 Climate

> All prices and numbers are our best guess for the prototype. We still need to test a real machine. Say this when you give numbers.

01 · THE PROBLEM

Burning is the cheapest way, so farmers burn

Farm land burned in 2026 (million rai)



Source: GISTDA 2026 (via Krungthep Turakij); Rocket Media Lab 2023–2025



North

The most burning, and the most lung cancer for each person.

Central & West

Corn burning went up 760% (Central). Rice burning went up 40 times (West).

East

Lung cancer is going up fastest, mostly in Sa Kaeo, where farmers grow a lot of corn.

Why farmers burn

It's the fastest and cheapest way to clean the field. A tractor costs money, and nobody buys the waste.

Who pays for it

Everyone: smoke, PM2.5, lung problems and cancer. Every 1 kg not burned means about **8 g less PM2.5** (our guess).

Key line: "Farmers know burning is bad. But it's the cheapest way. We want *not* burning to be the cheaper way."

02 · THE SOLUTION

KhumKha: one machine, three products



KhumKha (คุ้มค่า) means "worth it". Not burning should be worth more than burning. Size: 85 × 180 × 60 cm, about a water dispenser. The waste moves from the top to the bottom.

HEIGHT	PART	WHAT IT DOES
140–180 cm	Opening with a safety lid, hand guard, magnet, wet-meter, touch screen, QR scanner, camera	Takes in and checks the waste
100–140 cm	Cutter (about 3 kW) + 2 screens	Cuts it into small pieces
85–100 cm	Moving gate	Sends it to A, B or C
30–85 cm	A pellet press · C straight path · B hot press	Makes the product
0–30 cm	3 trays with scales	Weighs the product
Back	Dust fan and bag, power box, air holes	Safe, clean, cool

A · PELLETS

Animal food / fuel pellets

We get back	**85%**
Time	**~60 min / 100 kg**
Fee	**1.20 ฿/kg**

B · FUEL BLOCKS

Fuel instead of charcoal

We get back	**80%**
Time	**~75 min + 15 warm-up**
Fee	**1.50 ฿/kg**

C · SMALL PIECES

Compost · mushrooms

We get back	**~90%**
Time	**~15 min / 100 kg**
Fee	**0.40 ฿/kg**

How it works: 5 steps

1. **Ask in LINE**
   What waste, how much; bring it or ask for a pickup
2. **Village pickup**
   A village driver picks it up, like Grab
3. **Check**
   Weigh it, take a photo, check if it's dry
4. **Make it**
   A, B or C
5. **Farmer chooses**
   Sell to us, let us sell it, or take it home; receipt in LINE



Village station (~2 rai)

It's like a gas station: truck scale, drying area with a roof, machine room, solar roof, storage, shop and office, fire safety, parking. We start with **one machine in each village**.

> Limits: one product at a time · it cleans itself when it changes from fuel to animal food · extra blades + a technician every 200 hours

03 · PRICES & BUSINESS MODEL

How the money moves

WHAT	PRICE
We buy: corn waste, rice straw, fruit tree branches	**1.00 ฿/kg**
We buy: sugarcane leaves	**0.80 ฿/kg**
Wet: 15% or less / 15–20% / more than 20%	**full / -10% / no**
Fee to take the product home (A / B / C)	**1.20 / 1.50 / 0.40 ฿/kg**
You can ask for a pickup	**300 kg+ or a full truck**

Pickup fee (up to 10 km)

50 ฿ + 10 ฿/km + 0.10 ฿/kg

Selling to us: the farmer pays **30% of the sale or less**. Our fund pays the rest. The driver always gets **90%**. We keep **10%**.

1,000 kg of dry corn waste, 6 km
Sale 1,000 ฿ · pickup 210 ฿
Farmer gets 790 ฿
Driver 189 ฿ · we keep 21 ฿

300 kg, 6 km
Sale 300 ฿ · pickup 140 ฿ → farmer pays 90 ฿, our fund pays 50 ฿
Farmer gets 210 ฿
Driver 126 ฿

Business Model Canvas

Partners
Village groups · local government · university · mushroom farms, cow farms, longan dryers · drivers

Main work
Buy, make, sell · run LINE · take care of the machines

What we need
The machine · LINE system and data · drivers · our brand

What we give
Farmers: money and farm pickup · Buyers: material every month · Village: clean air

Relationship
LINE · clear receipts · public dashboard

How we reach them
LINE · village drying area · village leaders

Customers
Farmers who sell waste · buyers of our 6 products · drivers

Costs
Machine ~200k ฿ · electricity (solar helps) · buying waste · repairs · workers · pickup fund

Money in
Selling products (most money) · fees · 10% of pickups · later: sponsors, government, carbon credits

Price of one machine

~200,000 ฿ (150k–280k)

Cutter 25–40k · pellet press 25–45k · fuel block press 40–80k · body 15–25k · power box 10–20k · screen and camera 5–10k · sensors 4.5–11k · fan and other 7–13k · building it 15–30k

Pay it back (example)

400 kg a day × 25 days = 10 tons a month
× about 1.2 ฿ profit per kg = 12,000 ฿ a month
200,000 ÷ 12,000 = about **17 months**

04 · IMPACT

Four SDGs. Everyone gets something.

3
GOOD HEALTH
No burning → less smoke → fewer people with lung problems

8
GOOD JOBS
Farmers get money from waste. Village people get jobs as drivers.

12
NO WASTE
Crop waste becomes animal food, fuel and compost

13
CLIMATE
Less carbon from burning. We count it on the dashboard.

Also: SDG 7 (clean fuel + solar roof) and SDG 15 (fewer forest fires, waste goes back to the soil).

Farmers
Money from the waste they used to burn. No need to carry it. No fines. Clear receipts in LINE.

Village drivers
Extra money with their own truck. Always 90% of the full fee.

Buyers
Cheap material every month, instead of charcoal, wood or expensive animal food.

The village
Cleaner air, fewer sick people, local jobs.

Local government
A real way to stop burning, with data from every job.

Nature
Less smoke and carbon. Waste goes back to the soil as compost.

People
Money for farmers and drivers

Planet
Less smoke and carbon

Profit
Selling products + fees + 10% of pickups

We count every job
kg not burned · less PM2.5 (about 8 g/kg) · money for farmers · money for drivers · money from the fund. You can see it all on the dashboard.

Software (it works now)
LINE (sell, ask for pickup, receipts, data consent) · driver page · machine screen · dashboard · 3D model. It costs 0 ฿ a month now.



05 · WHAT TO SAY

Scripts

30 seconds · people walking past

"Do you know the smoke in the dry season?" *(wait for a nod)*
This year, Thailand burned **26 million rai** of farm land. That's a record. Farmers burn because it is the cheapest way. So we made **KhumKha**. In Thai, it means **"worth it"**. It turns crop waste into **animal food, fuel or compost**, and you use it with **LINE**. Farmers get money, drivers get jobs, and everyone gets cleaner air. We help **SDG 3, 8, 12 and 13**.
"Do you want to try? Scan this QR. It takes 10 seconds."

4 minutes · judges

PART	SHOW	SAY	TIME
Start	Boards	"We are Rai Mai Phao. We make crop waste worth more than burning it."	10 s
Problem	Board 1 map	"Thailand burned 26 million rai in 2026. That's a record. The North burned 16 million and has the most lung cancer. Farmers burn because it's the cheapest way."	30 s
Machine + demo	Board 2 + **give the iPad to the judge** (Show workflow, X-ray)	"This is KhumKha. It means 'worth it'. It cuts crop waste and makes animal food, fuel blocks or compost. It weighs it, checks it, and sends a receipt to LINE."	50 s
System	Board 3 station	"Each village has a station, like a gas station. Farmers ask in LINE. A village driver picks it up, like Grab. The farmer pays 30% or less for the pickup. Our fund pays the rest."	30 s
Money	iPad → Business Model	"We make money in three ways: we sell the products, we take a small fee, and we keep 10% of each pickup fee. One machine costs about 200,000 baht. We can pay it back in about 17 months."	40 s
SDGs	SDG boxes, board 2	"SDG 3: cleaner air. 8: jobs and money. 12: no waste. 13: less carbon. Good for people, nature and business."	20 s
End	Phone with LINE	"Do you want to try? Sell some waste in LINE. You get the receipt in a few seconds."	20 s

Always say

"These numbers are our best guess. Next, we want to test a real machine."

Not much time?

Go straight to the iPad demo → Business Model → SDGs.

Greeter

At the front of the booth: "Do you know the smoke in the dry season?"

Speaker

Next to the boards. Uses the script and points at each board.

Demo

Holds the iPad and phone. Lets people turn the model and try LINE.

Questions and easy answers

PROBLEM & IDEA

Why do farmers burn?
It's the fastest and cheapest way to clean the field. Ploughing costs money.

Why not stop it with a law?
There are laws, but farmers still burn if there's no cheaper way. We give them money, so they stop.

Who are your customers?
Farmers who sell waste, and people who buy our products: mushroom farms, cow farms, animal farms, factories.

MACHINE

What is it called?
KhumKha. It means "worth it" in Thai.

What does it make?
A: animal food or fuel pellets · B: fuel blocks · C: compost or mushroom material.

How fast is it?
About 15 min for 100 kg of C, 60 min for A, 75 min for B.

What if the waste is wet?
More than 20% wet: we can't take it. The farmer can dry it at the station.

What if it breaks?
We keep extra blades. A technician checks it every 200 hours.

Is it safe?
Yes: safety lid, hand guard, magnet, red stop button, dust fan.

Did you build it?
We made a full 3D design and working software. Next, we build a real one.

MONEY

How do you make money?
We sell the products, take a small fee, and keep 10% of each pickup fee.

How much is the machine?
About 200,000 ฿ (150k–280k).

When do you get the money back?
In about 17 months, if we use 400 kg a day. This is a guess.

Why 1 baht a kilo?
It's better than burning (0 baht), and we still make a profit.

Who pays the pickup fund?
Money from selling our products. Only for farmers who sell to us.

Why 30%?
So the pickup fee doesn't take most of a small farmer's money.

Very small pickups?
No. You need 300 kg or a full truck. Small farms can share a trip.

Competitors?
Big power plants buy waste, but only near the plant and in big amounts. We go to small farmers.

IMPACT, TECH & LAW

Which SDGs?
3, 8, 12, 13 (also 7 and 15).

How do you measure impact?
Every job saves kg not burned, less PM2.5 and money for people. See it on the dashboard.

Is 8 g/kg real?
It's our guess. We will check it with research.

Why LINE?
Farmers already use LINE every day. No new app.

Personal data?
Farmers say yes first. We keep only what we need. They can delete it.

Can you sell the animal food?
First, we use it on village farms. To sell it, we need a permit.

HARD QUESTIONS

Biggest risk?
We haven't tested the numbers. Next, we build one real machine.

What would you do with money?
Build one machine in a Chiang Rai village, test it, and fix our numbers.

What did you learn?
The problem is money, not knowledge. So we make the good choice the cheaper choice.

07 · BEFORE WE OPEN

Checklist & links

10 minutes before

- [ ] iPad on **/ipad/3d**: model loaded, English, full screen
- [ ] Computer screen on **/present**, press **F**
- [ ] Phone: try one sale in LINE and get the receipt
- [ ] Wi-Fi + phone hotspot
- [ ] Charge the iPad and phone · power strip
- [ ] Put the QR sign near the walkway
- [ ] Practise the 30-second and 4-minute versions
- [ ] Practise giving the iPad and opening Business Model

Links

Website	**sustainable-final.vercel.app**
iPad · 3D model	**/ipad/3d**
iPad · Business Model	**/ipad/bmc**
Presentation	**/present**
Prices	**/prices**
Dashboard	**/dashboard**
Machine screen	**/kiosk**

> The iPad pages are also in the Team menu on the website.

Numbers to say

26.0 M
rai burned, 2026

16.17 M
rai in the North

1 ฿/kg
we pay

≤ 30%
farmer pays for pickup

90%
goes to the driver

KhumKha
= "worth it"

~200k ฿
one machine

~17 mo
to pay it back

One sentence to remember: "We make *not* burning the cheaper choice. Farmers get money, drivers get work, buyers get cheap material, and everyone gets cleaner air."